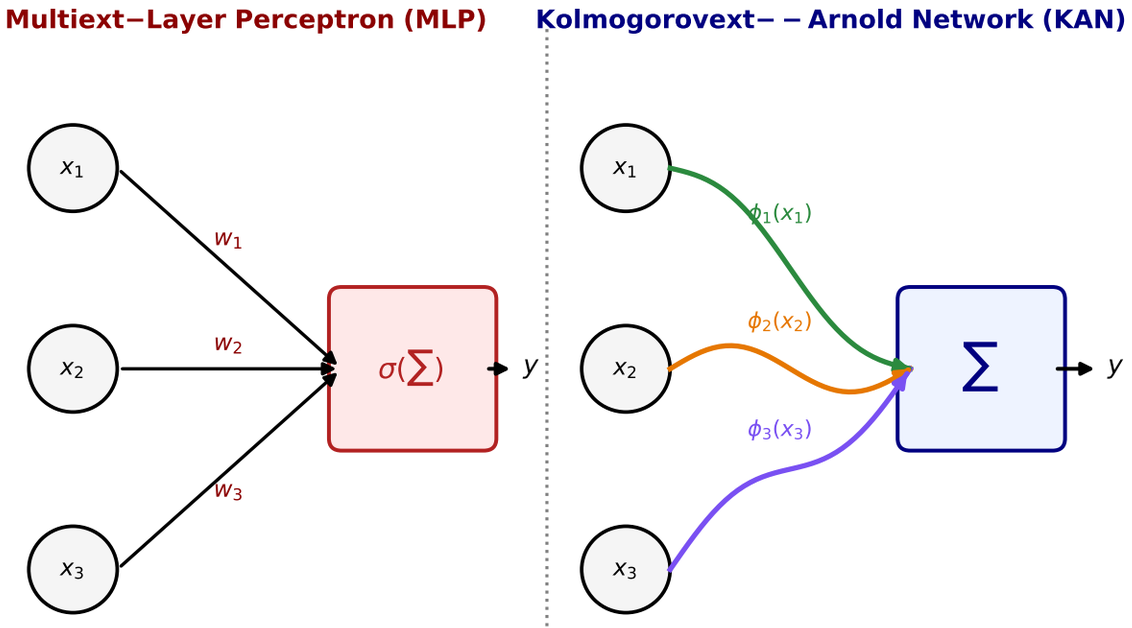
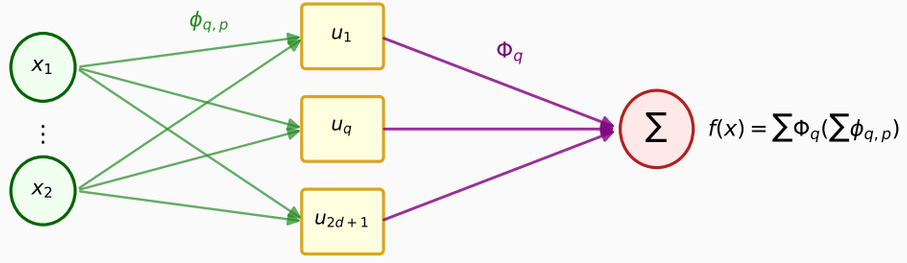
(a) Architectural Duality: MLP vs. KAN (Liu et al., 2024)

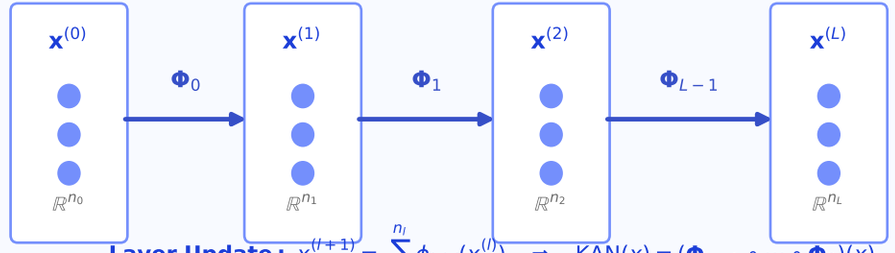


(b) From 2-Layer KA Theorem to Deep KAN ($L \geq 2$ Layers)

Classical 2ext−Layer KA Theorem (1957) : [do2d + 1o1]



Modern Deep KAN (Liu et al., 2024) : Arbitrary Depth $L \geq 2$ and Widths $[n_0, n_1, \ldots, n_L]$



Layer Update : $x_j^{(l+1)} = \sum_{i=1}^{n_l} \phi_{l,j,i}(x_i^{(l)}) \quad \Rightarrow \quad \mathrm{KAN}(x) = (\mathbf{\Phi}_{L-1} \circ \cdots \circ \mathbf{\Phi}_0)(x)$